

AETHER SERIES

Aether Dark

Dual-Voice Drone Synthesizer

A browser instrument with two analog-style voices, a shared drift LFO, dual 12 dB filters, modulated echo, multi-algorithm bloom, MIDI pitch control, and a virtual patchbay. This manual describes every control as implemented in the current software.

User Manual · Version 1.2 · Build 2026-09-25f

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1. What it is

Aether Dark is a two-voice drone synthesizer. There is no conventional keyboard on the panel. You hold tones with pitch knobs (or MIDI notes), then animate them with the Drift LFO, filters, echo, bloom and patch cables. It is an original Aether Series instrument: analog-modelled in the browser with the Web Audio API, not a sample player and not affiliated with any hardware manufacturer.

Each voice is a morphing oscillator (square / triangle / saw) mixed with a square sub-oscillator one or two octaves down, into a resonant 12 dB low-pass or high-pass filter. The two voices share one LFO rate but two independent waveform outputs. Effects sit after the voice mix.

Sound is generated in real time. Nothing is sampled from hardware. Results will differ between browsers because oscillators, filters and the convolver are the host implementation of Web Audio.

2. Quick start

1. Open **aether-dark.html** in a current desktop browser (Chrome, Firefox, Safari, Edge). For MIDI, serve the file over https or localhost — Web MIDI is blocked on most file:// pages and is not available in iPad Safari.
2. Click **POWER**. Browsers will not start audio until a gesture happens.
3. Load a factory patch such as Ambient Wash or Pitch-Linked Swarm, or leave the default panel and drag knobs vertically.
4. Stepped knobs (LFO waves, bloom algorithm) click to the next value, or drag with detents. Continuous knobs drag vertically. Double-finger scroll on a trackpad is not mapped.
5. Play pitches from a MIDI controller or the computer keys A W S E D F T G Y H U J K.

3. Signal flow

Voice I and Voice II run in parallel. Each path is:

Oscillator morph (X-Wave) + Sub oscillator → saturation → **12 dB LP/HP filter** → voice volume → mix bus.

The mix bus feeds a dry path and a **PT-style analog echo** (delay with feedback, high-pass and low-pass in the loop). Dry and wet echo sum into **Bloom** (convolution reverb with eight algorithms and a light chorus on the wet tail). Stereo left/right then go to the master output and analyser.

The Drift LFO does not live on the audio bus. It produces two bipolar control signals that the destination switches and patch cables can apply to pitch, X-Wave, cutoff and delay time. Patching a voice OUT into cutoff or pitch adds audio-rate modulation on top of that.

4. Transport, patches and toolbar

Control	What it does
POWER	Creates or resumes the AudioContext and starts the voice drones. Click again to suspend audio. The button reads POWER ON — DRONING when live.
Randomize	Scrambles LFO, voices, FX, noise and Evolve. Does not randomize cables.
Clear cables	Deletes every virtual patch cable and disconnects audio-rate routes.

Control	What it does
Patch menu	Factory patches under Factory; your saved sounds under User. Choosing a row loads knobs, switches, offset and cables.
Patch name	Label used when saving or exporting. Forty characters maximum.
Save	Writes the current panel (including cables) into browser localStorage under the typed name. Overwrites a user patch of the same name.
Delete	Removes the selected user patch only. Factory patches cannot be deleted.
Export	Downloads JSON: current patch plus the whole user bank. Type aether-dark-patch, version 1.
Import	Reads a previously exported JSON file. A params object becomes the current patch; a bank object is merged into User.

User patches persist in this browser profile under the key aether-dark-user-presets-v1. Clearing site data deletes them; keep exports.

5. Drift LFO and Evolve

One rate clock drives two shape outputs. Wave I and Wave II can be different shapes, including double- and quadruple-speed patterns, so they stay related in time without sharing a contour. LEDs under the knobs light when that output is positive.

Knobs and switches

Control	Range	Function
Wave I	Off ... Quad Square, 8 positions	Shape for output WAVE I. Click to cycle; drag uses detents. The readout under the knob is the live name.
Wave II	Off ... Quad Exp, 8 positions	Shape for output WAVE II. Phase-shifted and exponential variants live here.
Rate	0.02 Hz – 70 Hz, logarithmic	Shared cycle speed. At the slow end a cycle lasts about 50 seconds; at the fast end the LFO is a low audio rumble.
OSC I / OSC II dest.	WAVE I or WAVE II	Which LFO output is normalised to that voice's oscillator (pitch or X-Wave, chosen per voice).
FILTER I / FILTER II dest.	WAVE I or WAVE II	Which LFO output is normalised to that voice's cutoff.

Wave I shapes

#	Name	Behaviour
0	Off	Output held at zero. Destination switches still work but do nothing.
1	Triangle	Smooth bipolar rise and fall (implemented as a sine).
2	Square	Hard ± 1 . Good for trills, gated filters and stepped pitch.
3	Sawtooth	Falling ramp. Classic down-sweep on pitch or cutoff.
4	Roundtooth	Soft, compressed sine. Gentler than triangle.
5	2× Triangle	Two cycles per Rate period.
6	2× Exp Up/Dn	Two exponential peaks per period.
7	Quad Square	Four-step pattern of highs, lows and partials — a tiny sequencer.

Wave II shapes

#	Name	Behaviour
0	Off	Output held at zero.
1	Triangle 90°	Sine delayed a quarter cycle from Wave I triangle. Useful for stereo-ish opposite motion when I and II both use triangle family shapes.
2	Square 90°	Square delayed a quarter cycle.
3	Exp Saw	Exponential falling ramp. Snappy decays when mapped to pitch or cutoff.
4	Log Triangle	Logarithmic triangle. Spends more time near the peaks.
5	2× Saw	Two falling ramps per Rate period.
6	2× Pattern	Eight-step square pattern at double speed.
7	Quad Exp	Four exponential up/down lobes per period.

Depth amounts live on each voice (Osc LFO Depth, Filter LFO Depth) and on the echo (LFO Depth). Those knobs are bipolar attenuverters: 12 o'clock is off, clockwise follows the wave, counter-clockwise inverts it.

Evolve

A slow random walk that leans the existing voices. Depth 0 is off. Depth is 0 to 1. Time is the climate period (about 45 seconds to 8 minutes). Target chooses Tone (X-Wave and cutoffs), Weather (noise level), Motion (LFO and noise rates), or All. Irregular re-rolls extra offsets each LFO cycle. Loop wanders; Once settles. The Evolve CV jack outputs the walk.

Control	Range	Function
Time	45–480 s, log	How long a climate gesture takes.
Depth	0–1	How far parameters lean. Zero disables Evolve.
Target	Tone / Weather / Motion / All	Which family may drift.
Irregular	0–1	Cycle-to-cycle variation.
Loop / Once	toggle	Loop keeps aiming; Once sits after one walk.

6. Voice I and Voice II

The two voices are identical except Voice II also has Hard Sync, Pitch Link and a semitone Offset. Each voice is monophonic and always running while POWER is on (unless MIDI Gate has closed it).

Oscillator

Control	Range	Function
X-Wave	0–1	Morph across the current mode. Dark: square to triangle to saw. Dusk: square to triangle to sine to saw.
Osc Mix	0–1	Balance of the morphing oscillator versus the square sub. Fully clockwise is sub only.
Sub Osc	–1 or –2 octaves	Sub square sits one or two octaves below the main pitch.
LFO → Pitch / X-Wave	Pitch or X-Wave	Whether the normalised LFO moves frequency or the morph position.
Osc LFO Depth	–1 to +1	Attenuverter for that LFO assignment. Centre is off.
Pitch	27 Hz – 1.6 kHz, log	Base frequency of the voice. Tuner to the left shows note name, cents versus A440 and hertz. The tuner ignores LFO wobble so it stays readable.

Voice II only

Control	Range	Function
Hard Sync	Off / On	Forces Voice II toward a harmonic of Voice I. Adds edge and locked spectra. Approximate, not analogue hard-sync reset.
Pitch Link	Off / On	When on, Voice II frequency is Voice I transposed by Offset. The Voice II pitch knob is ignored while linked (unless MIDI is targeting Voice II alone).
Offset – / +	–24 to +24 semitones	Transpose applied to Voice II. With Pitch Link on, this is the interval above Voice I (default +7, a fifth). With Pitch Link off, it transposes Voice II's own pitch knob.

Filter

Control	Range	Function
LP / HP	Low-pass or high-pass	12 dB/octave state-variable style filter. High resonance will self-oscillate and can become a third tone.
Cutoff	40 Hz – 12 kHz, log	Corner frequency before LFO and cables.
Resonance	0–22	Q. Upper third of the throw is self-oscillation territory — turn Voice volume down first.
Filter LFO Depth	–1 to +1	Attenuverter from the chosen LFO wave onto cutoff. About ± 2.2 octaves at full throw, before cables.
Volume	0–1	Voice level into the mix bus. At zero the oscillators are hard-muted so they cannot leak under noise.

7. Noise

A looping white-noise buffer, coloured by a filter, with optional sample-and-hold. Destination Mix hears it dry-ish; Filter I or II shares that voice's cutoff, mode and resonance. LFO Depth, Rate multiplier and Phase Offset follow Wave I so the noise can pulse without a third LFO.

Control	Range	Function
Level	0–1	Noise amplitude.
Colour	White / Pink / Brown / S&H	Pink and Brown are low-passed white. S&H holds random levels at the noise LFO rate.
Dest	Off / Mix / Filter I / Filter II	Where the noise is heard.
LFO Depth	0–1	Wave I amplitude modulation of the noise.
LFO Rate x	0.25–4	Multiple of Drift Rate.
Phase Offset	0–360 deg	Shift versus Wave I.

8. Mixer, Echo, Bloom, Tape and master

Mixer

Control	Range	Function
Delay D/W	0–1	Echo mix. Fully left is dry-dominant; right adds repeats. Dry is never fully removed so drones do not disappear.
Reverb D/W	0–1	Bloom mix into the convolution tail.
Main Out	0–1	Master gain after bloom. Also mapped from MIDI CC 7 and CC 11.

Echo

A single delay line with filtering and feedback, in the spirit of bucket-brigade chips: short-to-medium times, darkening repeats, able to self-oscillate.

Control	Range	Function
Time	70–450 ms	Delay time. Also swept by the selected LFO when Echo LFO Depth is off-centre.
LFO Wave I / II	I or II	Which Drift output modulates Time.
LFO Depth	–1 to +1	How far Time moves. At high depth the repeats warble and can stretch past the delay buffer's comfort zone.
Repeats	0–0.92	Feedback. Near maximum the echo runs away into a third oscillator. Keep Main Out modest.

Bloom

Control	Range	Function
Mono / Stereo	Off = mono, On = stereo	Stereo switch on the panel. The wet path is already split through two slightly different chorus delays.
Algorithm	8 positions	Rebuilds the impulse. Click or drag the stepped knob. The name under the knob updates.
#	Algorithm	Character
0	ROOM 1	Very short, bright.
1	ROOM 2	Short to medium, bright.
2	ROOM 3	Short, warm.
3	PLATE 1	Medium-long, bright.
4	PLATE 2	Medium, bright.
5	HALL 1	Long, medium bright. Default.
6	HALL 2	Longest, warmest.
7	CH/HALL	Chorus motion plus a long warm tail.

Tape

Print path after Bloom: saturation, wow/flutter and wear. Print off skips the wet path. Wow is squared so low settings stay subtle, but it still moves the whole mix including noise.

Control	Range	Function
Print	Off / On	Engage tape colour.
Sat	0–1	Drive into a normalised tanh so loudness stays comparable.
Wow	0–1	Slow pitch drift. First third of the knob is mild.

Control	Range	Function
Wear	0–1	Dulls highs; hiss on desktop only.
REC PLAY STOP SAVE	transport	REC prints the master after rumble. PLAY is playback only. SAVE is 16-bit mono WAV. iPad Safari often cannot record.

Tape SAVE is a sketch. For a release, record the desktop browser into a DAW so Bloom stays stereo.

Master row

Control	Range	Function
Rumble HPF	18–160 Hz	Master high-pass (also on Tape rec). Default 22 Hz is almost flat.
FX / Dry	FX or Dry	Dry bypasses Echo and Bloom without forgetting the D/W knobs.
Dark / Dusk	Dark or Dusk	X-Wave law. Dusk inserts sine in the middle.
Meter	peak + RMS	Bar plus a white peak tick.

9. Virtual patch cables

The panel is semi-modular. Destination switches are the normalised routes. Cables add extra paths and do not break those normals. Light rings are outputs; dark rings are inputs.

Patching gestures

Drag from an output to an input, or click an output then click an input. Click a drawn cable to remove that connection. Alt-click or right-click a jack to pull every cable on that socket. One output may feed many inputs. Several cables into one input are summed. Cables are saved with user patches.

Jacks

Jack	Dir	Kind	Colour / notes
RATE	In	CV	LFO frequency FM from Wave I/II.
RESET	In	Gate	Rising edge of a Wave output restarts the LFO cycle.
WAVE I	Out	CV	Teal cable. Bipolar LFO I.
WAVE II	Out	CV	Violet cable. Bipolar LFO II.
PITCH (I / II)	In	CV	About ± 0.75 octave per full-scale CV, added after the panel LFO.
PWM (I / II)	In	CV	Offsets X-Wave (square $\leftrightarrow$ triangle $\leftrightarrow$ saw).
CUTOFF (I / II)	In	CV or audio	CV from an LFO, or audio-rate FM from a voice OUT.
OUT (I / II)	Out	Audio	Gold = Voice I, red = Voice II. Post-volume dry tap.
NOISE OUT	Out	Audio / CV	Noise tap.
EVOLVE CV	Out	CV	Slow climate walk.

Audio-rate routes (OUT $\rightarrow$ PITCH or CUTOFF) are real Web Audio connections into those AudioParams. They stay live even between animation frames. Self-patching a voice into its own cutoff with high resonance will scream — that is expected.

10. MIDI connection and modes

Aether Dark listens to every MIDI input the browser grants. There is no MIDI out, no clock, and no SysEx. Safari on iPhone and iPad does not implement the Web MIDI API; use a hardware computer keyboard or a desktop browser.

Connecting

On first load the page calls requestMIDIAccess. Accept the browser prompt. The device menu lists ports and refreshes when you plug something in. All listed inputs are heard; the menu is a roster, not a filter. The status dot is green when ports exist, amber when MIDI is present but empty, red when the API is missing or permission was denied.

Pitch target

Mode	Notes
Pitch → Both	Last note sets both voices to that frequency. If Pitch Link is on, Voice II is then transposed by Offset from Voice I (so Both + Link + Offset is the usual stacked interval).
Pitch → Voice I	Last note sets Voice I only. Voice II keeps its knob or link math.
Pitch → Voice II	Last note sets Voice II only and bypasses Pitch Link for that voice.
Split ≤C3 I / >C3 II	MIDI note 48 (C3) and below drive Voice I; notes above C3 drive Voice II. Each side remembers its last held key in the stack.

Hold and Gate

Hold (drone) is the native mode: a note sets pitch and the voice keeps sounding when you lift the key. **Gate** silences a voice when no keys remain assigned to it. Last-note priority with a hold stack: new notes push, releases pop, the most recent remaining note wins.

Pitch bend is ±2 semitones on the voices currently driven by MIDI. It does not move the pitch knobs. Program change 0–6 loads factory patches in this order: Ambient Wash, Hypnotic Pulse, Dark Sequence, Pitch-Linked Swarm, Self-Oscillating Filters, LFO Kick / Pump, Stereo Hall Drone.

11. MIDI implementation chart

Function: Dual-voice drone synthesizer Date: 2026-09-25
Model: Aether Dark Version: 1.2 (Build 25f) MIDI channels: OMNI in, no out

Function	Transmitted	Recognised	Remarks
Basic Channel Default	X	1–16 (OMNI)	All channels accepted
Basic Channel Changed	X	X	No channel filter
Mode Default	X	Mode 3 (omni off conceptually; all ch heard)	Last-note priority
Note On	X	O 0–127	Velocity ignored except 0 = off
Note Off	X	O	Hold mode keeps pitch
Velocity Note On	X	X	Not mapped
Aftertouch	X	X	
Pitch Bend	X	O 14-bit	±2 semitones
Control Change	X	O see table	
Program Change	X	O 0–n	Factory patches in menu order
System Exclusive	X	X	

Function	Transmitted	Recognised	Remarks
System Common	X	X	No MTC, song pos, song sel
System Real Time	X	X	No clock, start, stop
Aux Messages	X	X	
Notes			O = Yes X = No

Recognised control changes

CC	Controller	Destination	Scaling
1	Modulation wheel	Drift LFO Rate	Log 0.02–70 Hz
7	Channel volume	Main Out	Linear 0–1
11	Expression	Main Out	Linear 0–1 (same as 7)
71	Resonance / Sound 2	Voice I Resonance	Linear 0–22
72	Release / Sound 3	Echo Repeats	Linear 0–0.92
73	Attack / Sound 4	Echo Time	Linear 70–450 ms
74	Cutoff / Sound 5	Voice I Cutoff	Log 40 Hz–12 kHz
76	Vibrato rate / Sound 7	Voice II Resonance	Linear 0–22
77	Vibrato depth / Sound 8	Voice II Cutoff	Log 40 Hz–12 kHz
91	Reverb depth	Bloom mix	Linear 0–1
94	Detune / Effect 4	Echo mix	Linear 0–1

CCs write the panel state and move the corresponding knobs. There is no MIDI learn and no CC transmit. Do not expect NRPN, RPN, bank select or all-notes-off to do anything.

12. Akai MIDI Mix map

Choose Map: Akai MIDI Mix in the toolbar. Notes 1-26 on the Mix are buttons, not piano keys. Use a second keyboard or the on-screen C3-C4 keys for pitch. Generic CC mode ignores this table.

Knobs (CC)

CC	Destination
16 / 20 / 24 / 28	V.I cutoff / V.I res / V.II cutoff / V.II res
46 / 50 / 54 / 58	LFO Rate / Echo Time / Bloom algo / Tape Wear
17 / 21 / 25 / 29	V.I X-Wave / V.I osc depth / V.II X-Wave / V.II osc depth
47 / 51 / 55 / 59	Wave I / Echo LFO depth / Bloom mix / Tape Wow
18 / 22 / 26 / 30	V.I fil depth / V.I mix / V.II fil depth / V.II mix
48 / 52 / 56 / 60	Wave II / Echo repeats / Tape Sat / Echo mix
19 / 23 / 27 / 31	V.I vol / V.I pitch / V.II vol / V.II pitch
49 / 53 / 57 / 61 / 62	Offset / Echo mix / Main Out / Tape Print / Main Out

Buttons (note numbers)

Note	Action
1 / 7	Voice I LP-HP / Voice II LP-HP
4 / 10	V.I LFO Pitch-XWave / V.II LFO Pitch-XWave
3 / 6 / 9 / 12 / 15	LFO source toggles for osc I, fil I, osc II, fil II, echo
13 / 16 / 19 / 22	Tape Print / Pitch Link / Hard Sync / Hold-Gate
18 / 21 / 24	Tape REC / PLAY / STOP
25 / 26	Previous / next factory patch

13. Computer keyboard

A one-octave fallback when MIDI is absent. Ignored while typing in the patch-name field.

Key	Note	MIDI #	Key	Note	MIDI #
A	C3	48	Y	G#3	56
W	C#3	49	H	A3	57
S	D3	50	U	A#3	58
E	D#3	51	J	B3	59
D	E3	52	K	C4	60
F	F3	53	O	C#4	61
T	F#3	54	L	D4	62
G	G3	55	P	D#4	63

Keys use the same Hold / Gate and Pitch Target modes as MIDI. There is no octave shift on the computer keyboard.

14. Factory patches

Name	Starting point
Ambient Wash	Slow linked fifth, both low-pass, long Hall 2, gentle opposite filter motion.
Hypnotic Pulse	Square LFO on filters, octave link, plate bloom, pulsed delay.
Dark Sequence	Quad-square pitch pattern, hard sync, low growl, short room.
Pitch-Linked Swarm	Near-unison with opposite pitch LFO, open filters, chorus-hall.
Self-Oscillating Filters	Resonance at the top, voices almost inaudible, filters as oscillators.
LFO Kick / Pump	Saw LFO slams Voice I pitch and cutoff; Voice II as a ticking high hat-like layer. Gate not required — the LFO does the envelope.
Stereo Hall Drone	Very slow X-Wave motion, fifth, CH/HALL bloom, long echo.
North Wind	Pink noise into Filter I, low voices, long hall.
Gale / Wire Fence	White noise, high-pass howl, more motion.
Shoreline	Brown noise on the mix, slow filters, long repeats.
Shoreline · Breakers	Brown noise into Filter I with deep amplitude LFO.
Rain on Glass	White noise into Filter II, short bright room.

Name	Starting point
Thunder Bank	Brown noise into Filter I; noise cable into Voice II cutoff.
Night Insects	High thin voices, a little pink noise.
Deep Tide / Long Climate	Slow drones with Evolve.
Fog Bank / Chapel Fifth / Glacier / Embers	Atmosphere set. Embers uses S&H noise.

15. Ways to play it

Aether Dark rewards sitting still. Change one thing, listen for a full LFO cycle, then change another. The ideas below are starting points, not presets.

Drones and intervals

Pedal fifth. Pitch Link on, Offset +7. Both voices low-pass, cutoffs staggered. Wave I triangle very slow onto Filter I, Wave II 90° triangle onto Filter II so the two vowels breathe out of phase.

Just a beating unison. Pitch Link on, Offset 0. Raise Osc LFO Depth on Voice I a hair and invert Voice II. The swarm preset is this idea already.

Organ quartal stack. Offset +5 (fourth) or +12 (octave). X-Wave near triangle, mix a little sub at -2. Hall 2, delay almost off.

LFO as performer

One-thumb sequences. Wave I = Quad Square into OSC I pitch with modest depth. Rate around 2–4 Hz. Leave Voice II linked a fifth so the pattern is harmonised. Change Wave I to Quad Exp for slurred figures.

Sidechain without a compressor. Wave I saw into Filter I at high depth, cutoff low, resonance moderate. The filter closes on the ramp and the drone pumps. Kick / Pump factory patch is the aggressive version.

LFO FM. Patch WAVE II into RATE. Set Wave II to Exp Saw and Rate slow. The clock speeds up and slams back — useful under a looping echo.

Reset patterns. Patch WAVE I (square) into RESET while Wave II is a slow triangle on cutoff. The triangle never finishes; it keeps restarting into a bowed shape.

Filters as oscillators

Sine choir. Resonance full, voice oscillators mixed almost out, cutoff near the pitch you want. Two self-oscillating filters a fifth apart with tiny opposite LFO depths sound like tired sine waves. Add CH/HALL.

Audio-rate clang. Patch Voice II OUT into Voice I CUTOFF. Raise Voice I resonance. Move Voice II pitch through harmonics of Voice I. This is FM of the filter and will alias; that is part of the colour.

Echo and bloom as voices

Third oscillator. Repeats near maximum, Time 180–250 ms, a little LFO on Time. Drop dry voices so the delay is the instrument. Touch Rate and listen to the repeats stretch.

Aux feedback adjacent. Patch Voice I OUT to AUX IN with echo already up. You are stacking another send into the same loop. Keep Main Out low.

With MIDI

Hold a root, play a melody on top. Split mode: left hand below C3 parks Voice I, right hand above C3 moves Voice II. Offset off, Pitch Link off. Let bloom glue the two registers.

One-finger changing drones. Target Both, Hold, Pitch Link, Offset +7. Each key is a new fifth. Mod wheel (CC 1) becomes the only performance control you need after that.

Filter from a surface. Map a MIDI knob to CC 74 (Voice I cutoff) and CC 77 (Voice II cutoff) if your controller can send both. Play cutoffs, leave pitch parked.

Live discipline

Start every set with Main Out low. Self-oscillating filters plus max repeats plus Hall 2 will clip laptop speakers and surprise a front-of-house engineer. Change Rate last — it rewrites the meaning of every other modulation. When lost, Clear cables, load Ambient Wash, and bring one cable back.

16. Hosting, Safari and iPad notes

Web MIDI and reliable localStorage need a secure context (https or localhost). Opening the HTML file from disk works for audio in most desktop browsers after POWER, but MIDI may be denied and iPad Files-app open is unreliable.

iPad Safari will run the oscillators, knobs and patchbay. It will not list MIDI devices. Attach a hardware keyboard for A–K notes, or treat it as a pure drone panel. Backgrounding Safari suspends the AudioContext — tap POWER when you return. Export-as-download is flaky in iOS Safari; import usually works.

There is no installable wrapper in this build: no service worker, no Home Screen manifest. Adding the hosted page to the Home Screen is still a convenient bookmark.

Host the public build on GitHub Pages as index.html plus this PDF. On iPad open the https Pages URL in Safari, not the Files preview. Support: ko-fi.com/tribalvoice.

17. Specifications

Item	Value
Voices	2 always-on drones, monophonic each
Oscillators per voice	Saw + triangle + sine + square; Dark or Dusk morph; sub −1/−2
Filters	2 × biquad, 12 dB LP or HP, self-oscillating
LFO	0.02–70 Hz shared rate, 2 × 8 shapes (including Off)
Evolve	45–480 s random walk; Tone / Weather / Motion / All
Noise	White / Pink / Brown / S&H; Mix or Filter I / II
Echo	70–450 ms, filtered feedback, LFO on time
Bloom	8 convolution algorithms, optional chorus tail
Tape	Sat / Wow / Wear print; mono 16-bit WAV save
Master	Rumble HPF 18–160 Hz, FX dry bypass
Patch points	LFO, Evolve, both voices, noise
MIDI in	Notes, pitch bend ±2 st, Generic CC or MIDI Mix, program change
MIDI out	None
Tuning reference	A4 = 440 Hz
Host	Web Audio API in a single HTML file
Persistence	localStorage user bank + JSON import/export

Aether Dark is part of the Aether Series. It is original software. Panel architecture is a common dual-voice drone layout and is not a reproduction of any commercial hardware manual.